A VANISHING-CYCLE OBSTRUCTION TO SMOOTHING CALABI–YAU THREEFOLDS

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Abstract. A projective Calabi–Yau threefold whose singular points all admit local smoothings need not admit a global one. For threefolds whose singularities are ordinary double points and anticanonical cones over del Pezzo surfaces of degrees 6 and 7, we prove a necessity criterion: any one-parameter smoothing produces a rational homology class on the disjoint union of the singularity links that vanishes on a crepant resolution, lies in a canonical root subspace of $K^\perp$ at each del Pezzo cone point, and is nonzero at every node, at every degree-7 point, and at every degree-6 point smoothed through its one-parameter component. The engine is a period computation: the integral of the holomorphic 3-form over a vanishing cycle equals the local smoothing parameter times a unit, so a smoothed germ cannot have rationally trivial vanishing homology. In the purely nodal case the criterion recovers the necessity half of Friedman's relation for arbitrary one-parameter smoothings, without appeal to unobstructedness. The criterion is a finite linear-algebra test, and it obstructs: the Δ-regular anticanonical hypersurface of the unique doubly isolated Batyrev mirror pair, with 26 nodes, two degree-7 and two degree-6 cone points, and every germ locally smoothable, admits no smoothing. On a second hypersurface, with 14 nodes and one degree-7 point, the test is silent; we pose its smoothability, a genuinely mixed nodal and divisorial smoothing if it exists, as an open problem.

1. Introduction

1.1. **The problem.** Let X be a projective threefold over $\mathbb{C}$ with trivial dualizing sheaf and isolated singularities. When every singular germ of X admits a smoothing, one may ask whether X itself does. For ordinary double points the answer is classical: by Friedman's criterion [Fri86], simultaneous smoothing is governed by a relation among the exceptional curves of a small resolution in which every curve appears with nonzero coefficient; see also Rollenske–Thomas [RT09] for a topological formulation in all odd dimensions. Outside the nodal case much less is known. The local deformation theory of the simplest non-nodal germs, the anticanonical cones over del Pezzo surfaces, has been understood since Altmann [Alt95, Alt97], and their smoothing components have explicit geometric descriptions (Gross [Gro97, §5.5]; Petracci [Pet20, Pet19]). What has been missing is a global criterion that couples nodal and divisorial smoothing data.

The first three papers of this program [Wue26a, Wue26b, Wue26c] classified the isolated Gorenstein toric threefold germs arising on Batyrev hypersurfaces from unit-edge polygons, constructed global smoothings when a dual-edge-length invariant is at least two, and isolated a unique Batyrev mirror pair, associated with a 22-vertex reflexive 4-polytope $\Delta_{\mathbb{F}_1}$ and its polar, for which both members have only isolated singularities and the remaining smoothing question is genuinely mixed: one member carries a rigid cone singularity and is locally obstructed, while the other, denoted X° throughout, has 26 ordinary double points, two anticanonical cones

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over the del Pezzo surface of degree 7, and two over the del Pezzo surface of degree 6: thirty germs, every one of them locally smoothable. A purely nodal analysis cannot decide X°: the node relation matrix has two coloops, so any smoothing would have to recruit the divisorial deformation directions.

1.2. **Results.** Fix the following setting. X is a compact complex threefold with $\omega_X \cong \mathcal{O}_X$ and $h^1(\mathcal{O}_X) = 0$ whose singular points are ordinary double points and anticanonical cones over del Pezzo surfaces of degrees 6 and 7; $\pi\colon \widehat{X} \to X$ is a crepant resolution, small over the nodes and with an exceptional del Pezzo surface E_p over each cone point p. For a singular point p let L_p denote its link. A *one-parameter smoothing* of X is a flat proper analytic family $\mathcal{X} \to \Delta$ over a disk with $\mathcal{X}_0 \cong X$ and X_s smooth for all $s \neq 0$. Such a family induces at each singular point a smoothing of the local germ, with Milnor fiber M_p, and we set

$$R_p \ := \ \ker\big(H_2(L_p;\mathbb{Q}) \longrightarrow H_2(M_p;\mathbb{Q})\big),$$

the part of the link homology that bounds in the local Milnor fiber.

Theorem 1.1 (mixed necessity; Theorem 6.1)**.** *Suppose X as above admits a one-parameter smoothing. Then the space*

$$K(X) \ := \ \Big\{\kappa \in \ker\Big(\bigoplus_p H_2(L_p;\mathbb{Q}) \to H_2(\widehat{X};\mathbb{Q})\Big) \ : \ \kappa_p \in R_p \text{ at every cone point } p\Big\}$$

contains, for every node q, for every degree-7 cone point q, and for every degree-6 cone point q whose induced local smoothing lies in the one-parameter component of its miniversal base, a class κ with $\kappa_q \neq 0$. Here the maps are induced by the inclusions $L_p \subset X \smallsetminus \operatorname{Sing} X \subset \widehat{X}$.

The subspaces R_p are computed in closed form (Proposition 2.1): at a degree-7 point R_p is the unique line spanned by a (-2)-class of $K^\perp \subset H_2(E_p;\mathbb{Q})$, and at a degree-6 point it is the A_1 or the A_2 summand of the root lattice $K^\perp \cong A_1 \oplus A_2$, according to the component of the local miniversal base through which the family smooths the germ. These subspaces coincide with the lattices $\Xi(V)$ attached by Kuznetsov–Prokhorov [KP23, Theorem 1.9] to the del Pezzo threefolds V that sweep out the local smoothings.

Theorem 1.1 is a finite linear-algebra test on any concrete X, and on the distinguished hypersurface it obstructs.

Theorem 1.2 (Theorem 6.5)**.** *Let X° be a Δ-regular anticanonical hypersurface in the Gorenstein Fano toric fourfold associated with the polar of $\Delta_{\mathbb{F}_1}$. Then X° admits no one-parameter smoothing. Moreover no fiber of a small representative of the miniversal deformation of X° is smooth, and no formal deformation of X° meets all thirty local smoothing loci.*

Combined with the local rigidity of its mirror partner [Wue26a, Wue26c], both members of the unique doubly isolated mixed mirror pair are non-smoothable: one because a single germ admits no local smoothing, the other for a global reason invisible to every germ.

Two by-products deserve separate statement. First, the mechanism behind Theorem 1.1 is elementary and quantitative: the period of the holomorphic 3-form of the nearby fiber over the vanishing cycle of a node equals the local smoothing parameter times a unit (Lemma 3.1), and an analogous nonvanishing holds for the distinguished cycles of the degree-7 and one-parameter degree-6 smoothings (Lemma 3.3), where the decisive constant is an explicit period equal to $4\pi^2$ on a cycle we construct by hand. Consequently a smoothed germ cannot have rationally null-homologous vanishing homology, at any order of tangency of the family with the local discriminants. In the purely nodal case this yields:

Corollary 1.3 (Corollary 6.3)**.** *For a compact complex threefold with $\omega \cong \mathcal{O}$, $h^1(\mathcal{O}) = 0$ and only ordinary double points, any one-parameter smoothing forces, for every node q, the existence of a relation among the exceptional curve classes of a small resolution that is nonzero at q. This is the necessity half of Friedman's criterion, for arbitrary one-parameter smoothings, obtained without unobstructedness of the deformation space.*

Second, on X° the argument localizes the obstruction topologically:

Proposition 1.4 (Proposition 6.4)**.** *In any hypothetical one-parameter smoothing of X°, the vanishing 3-spheres of the two coloop nodes would be null-homologous in $H_3(X_s; \mathbb{Q})$.*

The two statements collide (Lemma 3.1 forbids exactly this), and that collision is the proof of Theorem 1.2. A complementary second proof, with a different period input, runs through a degree-7 point instead of the nodes (Section 6).

Finally, the criterion is sharp in the sense that it does not obstruct indiscriminately. Applying the same finite test to the anticanonical hypersurface X_{19} of a 19-vertex reflexive polytope from the companion paper [Wue26c] (fourteen nodes and one degree-7 cone point, again with two nodal coloops), the finite test is silent: no covered coordinate is forced to vanish (Section 7), and we ask:

Question 1.5. Is X_{19} smoothable? Any smoothing of X_{19} would be a simultaneous smoothing of nodal and del Pezzo cone singularities that no purely nodal mechanism can produce.

1.3. **Method.** The proof has three independent inputs. *Membership*: a surgery decomposition of the nearby fiber along the links, together with Mayer–Vietoris, places the boundary of every 3-cycle of X_s into the space $K(X)$ of Theorem 1.1; the computation of the subspaces R_p rests on the classification of del Pezzo threefolds of degrees 6 and 7 (Fujita [Fuj80]) and on an identification, apparently not available in the literature in the form needed, of the germ Milnor fiber with a del Pezzo threefold complement *as a manifold with boundary*, compatibly with the two natural circle fibrations of the link (Lemma 4.2). *Periods*: the vanishing-period computations of Section 3 convert nonvanishing of local smoothing parameters into nonvanishing of homology classes. *Finite input*: for X°, an exact integral computation on a fixed crepant resolution shows that on the relevant restricted kernels the coordinates of the two coloop nodes and of the degree-7 directions vanish identically (Section 5); all decisive matrix data are printed, and every computation is reproduced by the ancillary programs (Appendix A).

We emphasize what the argument does *not* use. It does not use the first-order deformation theory of X beyond motivation: the localization map, its image, and the comparison of coordinate systems on local deformation spaces never enter. It is insensitive to the order of tangency of the smoothing family with the local discriminants, the phenomenon that blocks first-order methods: the necessity statements of [Fri86, RT09] concern the first-order data of a deformation class, and their proofs are specific to nodes (one-dimensional local T^1, quadric exceptional geometry), whereas the periods above see the actual arc. Neither does it use unobstructedness, which is unavailable for the non-lci cone germs.

1.4. **Relation to prior work.** Friedman [Fri86] proved the nodal criterion; Rollenske–Thomas [RT09] gave a topological proof and the higher-dimensional statement, in both cases for first-order data, with the passage to actual smoothings supplied by unobstructedness (Kawamata, Tian, Ran) in the nodal case. Friedman–Laza [FL25, FL24] identify the first-order localization data of threefolds with isolated singularities in terms of link homology; our Section 5 consumes only the resolution-side avatar of that picture, in the form of an integral matrix. Gross [Gro97] determined the smoothing components of the del Pezzo cone germs; Altmann [Alt95, Alt97] computed their miniversal deformations; Petracci [Pet20, Pet19] and Kaloghiros–Petracci [KP21] give modern treatments, the latter computing the Betti numbers of the degree-6 Milnor fibers, in agreement with Section 2. The classification of the sweeping del Pezzo threefolds is due to Fujita [Fuj80]; we follow the caution recorded there (Supplementary notes) that the degree-6 case of Iskovskikh's earlier list omits $\mathbb{P}(T_{\mathbb{P}^2})$, and cite Fujita, with Kuznetsov–Prokhorov [KP23] as the modern reference. On the constructive side, multi-parameter deformations of toric varieties in a single character degree were built by Ilten–Vollmert [IV12]; the sufficiency counterpart of Theorem 1.1, in particular Question 1.5, is the subject of work in progress and is not addressed here.

Convention 1.6. dP_d denotes a del Pezzo *surface* of degree $d = K^2$; thus $\mathrm{dP}_6 = \mathrm{Bl}_3\,\mathbb{P}^2$ and $\mathrm{dP}_7 = \mathrm{Bl}_2\,\mathbb{P}^2$. A *del Pezzo threefold* is a smooth projective threefold V with $-K_V = 2H$ for an ample H; its degree is H^3. For a polygon or polytope, edges are *unit* if they have lattice length one. Homology and cohomology have $\mathbb{Q}$-coefficients unless stated otherwise. The anticanonical cone over a del Pezzo surface S is $C(S) = \operatorname{Spec}\bigoplus_{k\geq 0} H^0(S, -kK_S)$.

2. The germs: links, miniversal bases, Milnor fibers

Throughout this section $(Y, 0) = C(S)$ is the anticanonical cone over $S = \mathrm{dP}_6$ or dP_7, an isolated Gorenstein toric threefold singularity: it is the toric variety of the cone over the unit-edge reflexive hexagon, respectively pentagon, placed at height one. (The unit-edge polygon here is the *fan* polygon of S; its polar, the Newton polygon of $-K_S$, is not unit-edge for the pentagon.)

2.1. **Links.** The punctured cone is the total space of $\mathcal{O}_S(K_S)$ minus its zero section, so the link L is the unit circle bundle of $\mathcal{O}_S(K_S)$, and the homology Gysin sequence together with $H_1(S) = 0$ gives a canonical isomorphism

$$H_2(L;\mathbb{Q}) \;\cong\; K_S^{\perp} \;\subset\; H_2(S;\mathbb{Q}), \tag{1}$$

of rank 3 for dP_6 and 2 for dP_7. For a node, $L \cong S^2 \times S^3$ and $H_2(L;\mathbb{Q}) = \mathbb{Q}$. The lattice $K^{\perp}(\mathrm{dP}_6)$ is the root lattice $A_1 \oplus A_2$ (discriminant 6); $K^{\perp}(\mathrm{dP}_7)$ has discriminant 7 and contains exactly one pair $\pm$ of primitive classes of square -2. Both facts are elementary and are certified by the ancillary programs together with everything else in this section that is finite.

2.2. **Miniversal deformations.** By Altmann's theorems, the deformation theory of these germs is completely explicit. T^1 is concentrated in the Gorenstein character degree, of dimension $N - 3$ for an N-gon [Alt95, (6.5) and the subsequent Remark]; the versal family in that degree, constructed in [Alt97], is therefore a versal deformation of the affine cone, with base ring computed there:

$$\begin{array}{lll} \mathrm{dP}_7: & \mathbb{C}[s_1, s_2]/(s_1^2,\ s_1 s_2) & \text{[Alt97, (9.1)(v)]}, \\ \mathrm{dP}_6: & \mathbb{C}[s_1, s_2, s_3]/(s_1 s_3,\ s_2 s_3) & \text{[Alt97, (2.5)]}. \end{array}$$

The reduced components correspond to the maximal decompositions of the polygon into Minkowski sums of lattice polygons [Alt97, (2.5), (8.1)]: the pentagon decomposes in exactly one way, into a unit segment pair and a unimodular triangle, giving one reduced component, the line $\{s_1 = 0\}$; the hexagon decomposes as a sum of two triangles (the line $\{s_1 = s_2 = 0\}$) and as a sum of three segments (the plane $\{s_3 = 0\}$).

Three standard supplements, used later, deserve explicit statement. First, *minimality*: the Kodaira–Spencer map of Altmann's family is an isomorphism onto T^1 [Alt97, (6.2), (6.3)], so the versal family is miniversal. Second, *the germ*: T^1 and T^2 of the affine cone are finite-dimensional and supported at the vertex, and the deformation theory of the affine variety is identified with that of the analytic germ $(Y, 0)$ by Artin approximation [Art69]; the analytic miniversal deformation exists by Grauert [Gra72]; the base is the analytic germ of the ring displayed above. (The versality criterion used in [Alt97, (7.2)] is that of de Jong–van Straten [dJvS94, §4].) Third, *arcs*: a morphism from the germ of a disk to the miniversal base factors through the reduction and, since the analytic local ring of a disk is a domain, lies in a single reduced component; for the pentagon base this forces $s_1 \equiv 0$ along any arc, and for the hexagon base it forces the arc into the line or into the plane. We refer to the corresponding local smoothings as smoothings *through* the given component.

2.3. **Sweeps and Milnor fibers.** The smoothing components of these cones are cones swept by hyperplane sections, in the sense of Pinkham [Pin74, (4.2), (5.1), (5.5)]; see [PP15, Lemma 4.1, Proposition 4.2] for the statement in all dimensions in the pencil form used here. Concretely (Gross [Gro97, §5.5]; Petracci [Pet20, Proposition 2.2], [Pet19, Proposition 4.12]): the general fiber of the degree-7 smoothing is $V_7 \smallsetminus \bar{S}$ with $V_7 = \mathrm{Bl}_{pt}\,\mathbb{P}^3$; the general fiber of the hexagon's one-parameter (line) component is $(\mathbb{P}^1)^3 \smallsetminus \bar{S}$; and that of its two-parameter (plane) component is $W \smallsetminus \bar{S}$ with $W = \mathbb{P}(T_{\mathbb{P}^2})$ the divisor of bidegree $(1,1)$ in $\mathbb{P}^2 \times \mathbb{P}^2$. In each case V is a del Pezzo threefold of the same degree with $\bar{S} \in |H|$ a copy of S, embedded by $-K_S = H|_{\bar{S}}$; the classification of the possible V (Fujita [Fuj80, (5.8), (5.16)]) shows there are no others. For the record, the open Milnor fibers of the three channels have $(b_2, b_3) = (1,1)$, $(2,1)$ and $(1,2)$ respectively; the degree-6 values agree with [KP21, Proposition 3.5].

Proposition 2.1. *Let $M = V \smallsetminus \bar{S}$ be one of the three sweep complements. Under the identification* (1) *of H_2 of the link $L = \partial(\text{tube around } \bar{S})$ with $K_{\bar{S}}^{\perp}$,*

$$\ker\big(H_2(L;\mathbb{Q}) \to H_2(M;\mathbb{Q})\big) = \ker\big(K_{\bar{S}}^{\perp} \to H_2(V;\mathbb{Q})\big) = \begin{cases} \text{the } (-2)\text{-line} & V = V_7, \\ A_1 & V = (\mathbb{P}^1)^3, \\ A_2 & V = W, \end{cases}$$

where the map is the restriction to $K^{\perp}$ of the pushforward $H_2(S;\mathbb{Q}) \to H_2(V;\mathbb{Q})$. For a node, $\ker(H_2(L;\mathbb{Q}) \to H_2(M;\mathbb{Q})) = H_2(L;\mathbb{Q}) = \mathbb{Q}$.

Proof. Since $\bar{S}$ is an ample divisor with $H_1(\bar{S}) = 0$, the Thom isomorphism gives $H_3(V, M) \cong H_1(\bar{S}) = 0$, so $H_2(M) \hookrightarrow H_2(V)$, and the inclusion $L \subset M$ composed with this injection factors, up to homotopy through the tube, as the projection $L \to \bar{S}$ followed by $\bar{S} \subset V$; the two kernels therefore coincide. The pushforward is computed in the standard bases. For V_7: a smooth member of $|H| = |2H_{\mathbb{P}^3} - E|$ is the strict transform of a quadric through the blown-up point, so $\bar{S} \cong \mathrm{Bl}_2\,\mathbb{P}^2$ with $H_{\mathbb{P}^3}|_{\bar{S}} = 2\ell - e_1 - e_2$ and $E|_{\bar{S}} = \ell - e_1 - e_2$; the pushforward matrix has rows $(2,1), (1,1), (1,1)$ on (ℓ, e_1, e_2) and kernel $\langle e_1 - e_2 \rangle$, which is the (-2)-line. For $(\mathbb{P}^1)^3$: the three projections restrict to the conic pencils $|\ell - e_i|$, the rows are $(1,1,1)$ and the unit vectors, and the kernel is $\langle \ell - e_1 - e_2 - e_3 \rangle = A_1$. For W: the two contractions restrict to ℓ and $2\ell - e_1 - e_2 - e_3$, and the kernel is $\langle e_1 - e_2,\, e_2 - e_3 \rangle = A_2$. For the node, $M \simeq T^*S^3$ has $H_2 = 0$. □

Remark 2.2. The three kernels are the lattices $\Xi(V)$ of Kuznetsov–Prokhorov: [KP23, Theorem 1.9(i)] states $\mathrm{Cl}(V) \cong \Xi(V)^{\perp} \subset \mathrm{Cl}(\bar{S})$ with $\Xi(V)$ negative definite of rank $10 - d - \rho(V)$, giving A_1, A_1, A_2 for V_7, $(\mathbb{P}^1)^3$, W, an independent confirmation.

2.4. **Which sweep for which component.** The one-parameter (line) component of the hexagon base sweeps to $(\mathbb{P}^1)^3$ and the two-parameter (plane) component to W: this is stated in [Pet20, Proposition 2.2], follows from [Alt95, (7.1.5)] (the total spaces over the two components are the cones over $(\mathbb{P}^1)^3$ and over $\mathbb{P}^2 \times \mathbb{P}^2$, whose anticanonical hyperplane slice is W), and is confirmed numerically by the Betti numbers quoted above against [KP21, Proposition 3.5]. The results of Sections 3–6 are stated so that only the degree-7 channel and the fact that each degree-6 kernel is A_1 or A_2 are consumed by the main theorem; the matching enters only in the sharper form of Theorem 6.1(3).

3. Vanishing periods

This section contains the analytic mechanism. Throughout, $\mathcal{X} \to \Delta$ is a one-parameter smoothing of a compact complex threefold X with $\omega_X \cong \mathcal{O}_X$, $h^1(\mathcal{O}_X) = 0$ and isolated singularities of the types in Theorem 1.1.

3.1. **A relative holomorphic 3-form.** Flatness and the Gorenstein property of all fibers make $\mathcal{X} \to \Delta$ a Gorenstein morphism after shrinking the disk, so the relative dualizing sheaf $\omega_{\mathcal{X}/\Delta}$ is invertible and compatible with base change [Con00]. We first prove $c_1(\omega_{X_s}) = 0$ integrally. Choose the disjoint good representatives and exterior W of Section 4.1. Over a neighborhood of W the family is a smooth proper submersion with boundary and hence is C^∞-trivial. Since $\omega_X|_W \cong \mathcal{O}_W$, the restriction of $c_1(\omega_{\mathcal{X}/\Delta})$ to the corresponding exterior in X_s vanishes. On each Milnor piece M_p, it also vanishes: the invertible sheaf $\omega_{\mathcal{X}/\Delta}$ is trivial on a sufficiently small total-space neighborhood of $(p, 0)$.

For every boundary link in this paper, $H^1(L_p; \mathbb{Z}) = 0$. This is immediate for the nodal link $S^2 \times S^3$. For a cone point, L_p is the circle bundle of $\mathcal{O}_S(K_S)$ over the simply connected del Pezzo surface S; the homotopy sequence shows that $\pi_1(L_p) = 0$ because its connecting map $\pi_2(S) \to \pi_1(S^1) = \mathbb{Z}$ is evaluation against the primitive Euler class K_S and hence is surjective. The Mayer–Vietoris sequence for $X_s = W \cup \bigsqcup_p M_p$ therefore injects

$$H^2(X_s; \mathbb{Z}) \longhookrightarrow H^2(W; \mathbb{Z}) \oplus \bigoplus_p H^2(M_p; \mathbb{Z}).$$

The preceding local vanishings give $c_1(\omega_{X_s}) = 0$. Semicontinuity gives $h^1(\mathcal{O}_{X_s}) = 0$; the exponential sequence then embeds $\operatorname{Pic}(X_s)$ into $H^2(X_s; \mathbb{Z})$ and forces $\omega_{X_s} \cong \mathcal{O}_{X_s}$. (The hypothesis $h^1(\mathcal{O}_X) = 0$ cannot be dropped: the cone over a smooth quadric surface in $\mathbb{P}^4$ is a one-node threefold smoothed by the family of quadric threefolds, and the vanishing cycle of that smoothing *is* null-homologous; the example is excluded here exactly because its ω is nontrivial.) Constants and upper semicontinuity give $h^0(\mathcal{O}_{X_s}) = 1$ after shrinking, hence $h^0(\omega_{X_s}) = 1$. By Grauert's base-change theorem [Gra60], $\pi_*\omega_{\mathcal{X}/\Delta}$ is then a line bundle on the shrunk disk; we fix a trivializing section Ω, whose restriction Ω_s is a nowhere-vanishing holomorphic 3-form on X_s for $s \neq 0$ and a generator of ω_X near each singular point at $s = 0$.

3.2. **Nodes.**

Lemma 3.1. *Let p be an ordinary double point of X. Then the vanishing-cycle class $\delta_p(s) \in H_3(X_s; \mathbb{Q})$ is nonzero for all small $s \neq 0$.*

Proof. The germ (X, p) is $\{xy - zw = 0\} \subset (\mathbb{C}^4, 0)$, with miniversal deformation $\{xy - zw = u\}$ over the u-line [KS72]. The family induces a classifying map $s \mapsto \tilde{u}(s)$, holomorphic with $\tilde{u}(0) = 0$ (only its nonvanishing at each fixed $s \neq 0$ is used, which is independent of choices), and X_s is smooth near p exactly when $\tilde{u}(s) \neq 0$; the hypothesis gives $\tilde{u}(s) \neq 0$ for all small

$s \neq 0$. The pullback total space is the hypersurface $\{xy - zw = \tilde{u}(s)\}$, Gorenstein even at its singular points, and adjunction exhibits $\omega_{\mathcal{X}/\Delta}$ near p as generated by the relative Gelfand–Leray form $\mathrm{GL}_u = \mathrm{Res}\big[dx\,dy\,dz\,dw/(xy - zw - u)\big]$. Hence $\Omega = g \cdot \mathrm{GL}$ with g holomorphic on the total-space germ and $g(p, 0) \neq 0$; extend g to an ambient holomorphic function so that $g(0, s)$ is defined.

After the linear change putting $xy - zw$ into the form $\sum_{i=1}^{4} v_i^2$, the vanishing cycle over u is the real slice $\delta_u = \sqrt{u} \cdot S^3$, whose class is single-valued: in fiber dimension three the Picard–Lefschetz transvection fixes δ (as $\langle \delta, \delta \rangle = 0$), and the antipodal ambiguity of $\sqrt{u}$ acts on S^3 with degree $+1$. Under $v \mapsto \lambda v$ one has $u \mapsto \lambda^2 u$ and $\mathrm{GL} \mapsto \lambda^2 \mathrm{GL}$, so the local period is exactly linear:

$$\int_{\delta_u} \mathrm{GL}_u \;=\; c_0\, u, \qquad c_0 = \int_{S^3} \frac{dv_1\, dv_2\, dv_3}{2v_4} = \pi^2 \neq 0$$

up to orientation conventions. For the twisted form, expanding the ambient extension $g = g(p, s) + O(|v|)$ and using $|v| = |u|^{1/2}$ on δ_u gives

$$F(u, s) := \int_{\delta_u} g \cdot \mathrm{GL}_u = u\,\big(c_0\, g(p, s) + O(|u|^{1/2})\big),$$

so $F(u, s) = u \cdot (\text{unit})$ near $(0, 0)$; here u, s are independent until the substitution $u = \tilde{u}(s)$.

Suppose $\delta_p(s_0) = 0$ in $H_3(X_{s_0}; \mathbb{Q})$ for some small $s_0 \neq 0$. A nonzero integer multiple of the cycle then bounds, and Ω_{s_0} is closed, so $\int_{\delta_p(s_0)} \Omega_{s_0} = 0$. But this period equals $F(\tilde{u}(s_0), s_0) = \tilde{u}(s_0) \cdot (\text{unit}) \neq 0$, a contradiction. □

3.3. **Del Pezzo cone points.** For the cone germs the same strategy applies, with the local model family replaced by the sweep and the explicit period computed on the sweep complement. Both relevant complements admit a uniform toric description. Let V be $(\mathbb{P}^1)^3$ or $\mathrm{Bl}_{pt}\, \mathbb{P}^3$ with its fan in $N \cong \mathbb{Z}^3$, and let $m \in M$ be the character $(1, 1, 1)$, respectively $(1, 1, -1)$ in the basis where the fan of $\mathrm{Bl}_{pt}\, \mathbb{P}^3$ has rays e_1, e_2, e_3, $r = -e_1 - e_2 - e_3$ and $e_1 + e_2 + e_3$. In both cases $\langle m, v \rangle = \pm 1$ on every ray, the closure $\bar{S}$ of $\{\chi^m = 1\}$ is a copy of the appropriate del Pezzo surface with $[\bar{S}] = -\frac{1}{2} K_V$, and

$$\Omega_M = \frac{w}{(w-1)^2}\, \frac{dx}{x} \wedge \frac{dy}{y} \wedge \frac{dz}{z}, \qquad w = \chi^m, \tag{2}$$

is the generator of $\omega(V \smallsetminus \bar{S}) \cong \mathcal{O}$, unique up to scale: the order of (2) along a boundary divisor D_v is $\langle m, v \rangle - 2 \min(\langle m, v \rangle, 0) - 1 = 0$, and along $\bar{S}$ it is -2. Here uniqueness is algebraic. If $f \in \mathcal{O}(V \smallsetminus \bar{S})^\times$, then $\mathrm{div}_V(f) = n\bar{S}$ for some $n \in \mathbb{Z}$; since $\mathrm{Pic}(V)$ is torsion-free and $[\bar{S}] \neq 0$, one has $n = 0$, and a rational function without zeros or poles on the projective variety V is constant. (These statements, the slice combinatorics below, and the constancy claim in the next proof are certified by the ancillary programs.)

Lemma 3.2. *In each of the two complements $M = V \smallsetminus \bar{S}$ above there is a closed 3-cycle $T \subset M$ with $\int_T \Omega_M = \pm 4\pi^2$. Consequently $[\Omega_M] \neq 0$ in $H^3(M; \mathbb{C})$, and since $b_3(M) = 1$, the period pairing $H_3(M; \mathbb{Q}) \to \mathbb{C}$, $\nu \mapsto \int_\nu \Omega_M$, is injective.*

Proof. Choose an edge $\langle v_+, v_- \rangle$ of the fan with $\langle m, v_\pm \rangle = \pm 1$ and an adjacent maximal cone $\langle v_+, v_-, n_0 \rangle$ with $\gamma := \langle m, n_0 \rangle = +1$; such pairs exist in both cases, any admissible choice works, and the transitions printed below arise from $(v_+, v_-, n_0; n_1) = (e_1, -e_2, e_3; -e_3)$ for $(\mathbb{P}^1)^3$ and $(e_1, r, e_2; e_3)$ for $\mathrm{Bl}_{pt}\, \mathbb{P}^3$. In the unimodular chart with coordinates (p, q, z)

dual to (v_+, v_-, n_0) one has $w = pq^{-1}z^\gamma$, $\bar{S} = \{t = 0\}$ with $t = q - pz^\gamma$, and $\Omega_M = \pm z^{\gamma-1}\,dp \wedge dq \wedge dz/t^2$. Fix $\varepsilon > 0$, $R > 1$ and set

$$T_1 = \{p = 0,\ q = \varepsilon R^\gamma e^{i\psi},\ |z| \le R\},$$
$$T_2 = \{p = r\varepsilon e^{i\theta},\ q = -(1-r)\varepsilon R^\gamma e^{i(\theta+\gamma\varphi)},\ z = Re^{i\varphi}\}, \qquad r \in [0,1],$$
$$T_3 = \{p' = \varepsilon R^c e^{i\psi},\ q' = 0,\ |z'| \le 1/R\} \quad \text{in the chart at the opposite corner } n_1,$$

where (p', q', z') are the dual coordinates of (v_+, v_-, n_1) and c is the z-exponent of the transition monomial for p' ($c = 0$ for $(\mathbb{P}^1)^3$, where the transition is $(p, q, 1/z)$, and $c = -1$ for $\mathrm{Bl}_{pt}\,\mathbb{P}^3$, where it is $(p/z,\ q/z,\ 1/z)$). The corner-chart cap is necessary: in the $\mathrm{Bl}_{pt}\,\mathbb{P}^3$ case the naive closure of $\{|p| = \varepsilon, q = 0, |z| \ge R\}$ collapses onto the torus-fixed point of the corner chart, which *lies on* $\bar{S} = \{p' = q'z'\}$; the cap T_3 instead keeps $|p' - q'z'| = \varepsilon R^c > 0$. On T_2, $t = -\varepsilon R^\gamma e^{i(\theta+\gamma\varphi)}$ has constant modulus; the $r = 0$ boundary is the boundary torus of T_1 and the $r = 1$ boundary maps under the transition onto ∂T_3, so with orientations of the caps chosen to cancel ∂T_2, $T = T_1 + T_2 + T_3$ is a closed 3-cycle in M.

The pullback of Ω_M to T_1 and T_3 vanishes identically ($dp \mapsto 0$, respectively $dq' \mapsto 0$, while Ω_M is regular there), and on T_2 the pullback of $z^{\gamma-1}dp \wedge dq \wedge dz/t^2$ is the *constant* form $dr \wedge d\theta \wedge d\varphi$: writing it out, the complex Jacobian of the parametrization equals $t^2 z^{1-\gamma}$ exactly. Hence $\int_T \Omega_M = \int_0^1\int_0^{2\pi}\int_0^{2\pi} dr\,d\theta\,d\varphi = 4\pi^2$ up to the orientation sign.

Finally $b_3(M) = 1$: from the pair sequence of (V, M) and $H^3(V) = 0$, $b_3(M)$ is the corank of the restriction $H^2(V;\mathbb{Q}) \to H^2(\bar{S};\mathbb{Q})$, which is one in both cases. □

Lemma 3.3. *Let p be a singular point of X whose germ is the anticanonical cone over* dP_7*, or over* dP_6 *with the induced local smoothing through the one-parameter component. Then $H_3(M_p;\mathbb{Q})$ ($\cong \mathbb{Q}$) $\to H_3(X_s;\mathbb{Q})$ is injective for all small $s \neq 0$.*

Proof. By Section 2.2 the induced arc lies in a single reduced component c of the miniversal base of the germ, off the local discriminant for $s \neq 0$, and by Section 2.3 the Milnor fiber is the corresponding sweep complement, which for the two channels in the statement is one of the complements of Lemma 3.2. The component carries Pinkham's $\mathbb{C}^*$-equivariant swept family with parameters of weight one (Section 2.2: T^1 sits in the Gorenstein degree); let Ω^{loc} be an equivariant algebraic generator of its relative dualizing sheaf, homogeneous of some integral weight a. To construct it, work in the graded local ring at the vertex. The relative dualizing module is invertible, and graded Nakayama selects from any algebraic local generator a homogeneous component that still generates. Its $\mathbb{C}^*$-saturation contains the entire fiber $b = 1$: every point of that fiber tends to the vertex when the positive-weight cone coordinates and the weight-one parameter are scaled to zero. Thus its restriction to $b = 1$ is an algebraic generator of ω_M, hence a constant multiple of Ω_M by the unit calculation following (2); rescale it so that $\Omega^{\mathrm{loc}}|_{b=1} = \Omega_M$. For a horizontal family ν of classes in $H_3(M;\mathbb{Q})$,

$$\int_{\nu(b)} \Omega_b^{\mathrm{loc}} \;=\; c(\nu)\cdot\chi(b),$$

with χ the weight-a monomial in the component parameter and $c(\nu) = \int_\nu \Omega_M$: single-valuedness holds because the $\mathbb{C}^*$-action is a group action on the total space with parameter weight one, so transport once around the base circle is the identity; and $c(\cdot)$ is linear. Writing $\Omega = g \cdot \Omega^{\mathrm{loc}}$ near p with g a unit as in Section 3.1, let $b = b(s)$ be the actual parameter of the component. The cycle obtained from the fixed compact cycle of Lemma 3.2 by the $\mathbb{C}^*$-action lies in the local Milnor piece and converges uniformly to p, because all cone coordinates have positive weights. Hence $g|_{\nu(s)} = g(p,0) + o(1)$ uniformly, while homogeneity factors the same monomial $\chi(b(s))$ from the main term and the error. The global period is therefore

$$\int_{\nu(s)} \Omega_s = \chi(b(s))\big(c(\nu)\, g(p,0) + o(1)\big).$$

If $\nu \neq 0$ in $H_3(M;\mathbb{Q})$ had $\nu_* = 0$ in $H_3(X_s;\mathbb{Q})$ for arbitrarily small $s \neq 0$, then a rational multiple of $\nu(s)$ would bound and the closed form Ω_s would have zero period, while $\chi(b(s)) \neq 0$ off the discriminant; letting $s \to 0$ along such values gives $c(\nu)\, g(p,0) = 0$ and, since $g(p,0) \neq 0$, $c(\nu) = 0$, contradicting the injectivity of the period pairing in Lemma 3.2. (That $H_3(M;\mathbb{Q})$ has rank one is what converts this class-by-class exclusion into injectivity for all small s; a higher-rank channel would require a uniform argument.) ☐

4. Surgery membership

4.1. **The decomposition.** Let $p_1, \dots, p_r$ be the singular points of X, with disjoint closed cone neighborhoods U_p chosen inside Milnor balls, and $W = X \setminus \bigcup_p \operatorname{int} U_p$, so $\partial W = \bigsqcup_p L_p$. Good representatives exist for arbitrary isolated singularities (Greuel [Gre20, Definition 2.12, Lemma 2.13, Theorem 2.15]), and for small s the family is smooth, hence C^∞-trivial, over a neighborhood of W; therefore

$$X_s \;\cong\; W \;\cup_{\bigsqcup L_p} \bigsqcup_p M_p, \qquad \widehat{X} \;\cong\; W \;\cup_{\bigsqcup L_p} \bigsqcup_p N_p,$$

where M_p is the Milnor fiber of the induced local smoothing and N_p a neighborhood of the exceptional set of the fixed crepant resolution. Both gluings are along the same links.

Lemma 4.1 (membership). *For every $\gamma \in H_3(X_s;\mathbb{Q})$, the Mayer–Vietoris boundary $\partial\gamma \in \bigoplus_p H_2(L_p;\mathbb{Q})$ lies in the space $K(X)$ of Theorem 1.1: its components die in every $H_2(M_p;\mathbb{Q})$, and their sum dies in $H_2(W;\mathbb{Q})$, hence in $H_2(\widehat{X};\mathbb{Q})$. Moreover, at a node q the L_q-component of $\partial\gamma$ is a fixed nonzero multiple of $(\gamma\cdot\delta_q)$ times the generator, and at a cone point p the boundary map identifies $H_3(M_p, L_p;\mathbb{Q}) \xrightarrow{\sim} R_p$.*

Proof. The first statement is Mayer–Vietoris exactness for $X_s = W \cup \bigsqcup M_p$, together with $W \subset \widehat{X}$. At a node, $M \simeq T^*S^3$: $H_3(M, L;\mathbb{Q}) \cong H^3(M;\mathbb{Q}) = \mathbb{Q}$ is generated by a cotangent fiber F with ∂F the S^2-generator of $H_2(S^2 \times S^3)$ and $F \cdot \delta = 1$; excision identifies the L_q-component of $\partial\gamma$ with ∂ of the image of γ in $H_3(M_q, L_q)$, which equals $(\gamma\cdot\delta_q)F$ by the perfect pairing. At a cone point, exactness of the pair sequence makes the image of $\partial\colon H_3(M, L;\mathbb{Q}) \to H_2(L;\mathbb{Q})$ equal to $\ker(H_2(L;\mathbb{Q}) \to H_2(M;\mathbb{Q})) = R_p$, while Lefschetz duality gives $\dim H_3(M, L;\mathbb{Q}) = b_3(M)$, which by Section 2.3 equals 1, 1, 2 in the three channels, the rank of R_p computed in Proposition 2.1. A surjection between spaces of equal finite dimension is an isomorphism, so ∂ is injective with image R_p; equivalently $H_3(M) \to H_3(M, L)$ is zero, $H_3(L) \to H_3(M)$ is surjective, and the skew intersection form of $H_3(M;\mathbb{Q})$ vanishes. ☐

4.2. **Boundary compatibility.** Lemma 4.1 uses the subspaces $R_p \subset H_2(L_p)$ through the *sweep-side* description of Proposition 2.1 (the unit normal circle bundle of $\bar{S} \subset V$), while the map to $H_2(\widehat{X})$ and all coordinates in Section 5 use the *resolution-side* description (the Seifert projection $L_p \to E_p$ of the cone link). These are a priori different identifications of $H_2(L_p;\mathbb{Q})$ with $K^\perp$, and an arbitrary boundary diffeomorphism need not preserve either marking. Their compatibility is therefore a statement, not a formality. It appears not to be available in the literature: the identification of the germ Milnor fiber with the sweep complement is standard as *open* manifolds ([PP15, Proposition 4.2]), but the manifold-with-boundary form, compatible with the canonical link identification, is what the following lemma supplies.

Lemma 4.2 (boundary compatibility)**.** *Let (X,p) be analytically isomorphic to the anticanonical cone over the del Pezzo surface S, smoothed by a one-parameter family through the component c with sweep $V \supset \bar{S}$. Then the boundary identification $\partial M_p \cong L_p$ arising from the surgery decomposition can be chosen so that the sweep-side projection $H_2(L_p;\mathbb{Q}) \to H_2(\bar{S};\mathbb{Q})$ and the Seifert projection $H_2(L_p;\mathbb{Q}) \to H_2(E_p;\mathbb{Q})$ agree up to an isomorphism of anticanonically polarized surfaces $(E_p, -K) \cong (\bar{S}, -K)$. In particular the comparison is a lattice isometry, and R_p, read in resolution-side coordinates, is the root subspace of Proposition 2.1.*

Proof. Let $\mathcal{Y}_c \to B_c$ be the $\mathbb{C}^*$-equivariant Altmann–Pinkham family on the reduced component c, and let $\overline{\mathcal{Y}}_c \to B_c$ be its projective sweep compactification. All component parameters have weight one, all affine cone coordinates have positive weights, and the divisor at infinity is the fixed family $S \times B_c$: negative weight does not change the top graded piece. Choose an equivariant affine embedding, write $w_1, \dots, w_N > 0$ for the coordinate weights and, for a common multiple d of them, use the weighted radius

$$\rho(z) = \left(\sum_{i=1}^{N} |z_i|^{2d/w_i} \right)^{1/(2d)}, \qquad \rho(r \cdot z) = r\rho(z) \quad (r > 0).$$

Fix a sufficiently small weighted Milnor radius ε and a small sphere of radius η in B_c. Every sufficiently small parameter $\beta \in c$ off the discriminant is uniquely $\beta = tb$ with $t > 0$ and $\|b\| = \eta$. The discriminant is homogeneous, so b is also off it. The action by t^{-1} gives a diffeomorphism of pairs

$$(\mathcal{Y}_\beta \cap \{\rho \le \varepsilon\}, \mathcal{Y}_\beta \cap \{\rho = \varepsilon\}) \;\cong\; (K_{t,b}, \Sigma_{t,b}), \quad K_{t,b} := \mathcal{Y}_b \cap \{\rho \le \varepsilon/t\}. \tag{3}$$

The pair on the left is the actual compact Milnor fiber and its boundary, not merely its interior.

We next identify the pair on the right with the sweep compact core. Near the fixed divisor $S \subset \overline{\mathcal{Y}}_b$, choose a normal coordinate q, and pass to the real oriented blowup along S. An affine coordinate of weight w_i has the form

$$z_i = q^{-w_i}\big(a_i(y,b) + O(q)\big),$$

where $y \in S$ and the leading coefficients do not vanish simultaneously. Consequently

$$\rho = |q|^{-1} A(y, \arg q, |q|, b) \tag{4}$$

with A smooth and strictly positive on the oriented boundary. For t small, the equation $\rho = \varepsilon/t$ therefore has a unique solution for $|q|$ and is a smooth radial graph over the unit normal circle bundle of S. Normal radial flow identifies this graph with the boundary of a fixed tube of S, preserving the projection to S, and extends across the intervening collar. Thus (3) continues to a diffeomorphism of pairs

$$(M_p, \partial M_p) \;\cong\; \big(\overline{\mathcal{Y}}_b \smallsetminus \operatorname{int}\operatorname{Tube}(S), \partial\operatorname{Tube}(S)\big)$$

that carries the actual boundary inclusion to the tube-boundary inclusion.

By the sweep description of Section 2.3, the compactified smooth pair $(\overline{\mathcal{Y}}_b, S)$ is the corresponding polarized sweep pair $(V, \bar{S})$; along a connected smooth channel the pair identifications may be transported by Ehresmann with the trivializing vector field chosen to vanish on the fixed divisor S.

It remains to identify the marking of that boundary. Apply the same construction to the radial family $\mathcal{Y}_{\tau b}$ for $0 < \tau \le t$. Formula (4) shows, on the real oriented blowup, that the rescaled boundary graphs extend smoothly to $\tau = 0$. Their limiting projection sends a nonzero point of the central cone to its projective class in S, while the normal circle is its S^1-orbit. It is therefore exactly the Seifert projection $L_p \to S$ of the cone link. We choose the boundary trivialization in the surgery decomposition to be this radial transport. The sweep-side tube projection and the resolution-side Seifert projection then induce the same map on H_2, after the polarized identification $(E_p, -K) \cong (S, -K) \cong (\bar{S}, -K)$.

Finally the A_1 and A_2 summands for dP_6 and the unique (-2)-line for dP_7 are intrinsic to the polarized lattice, so they correspond. Since the displayed diffeomorphism is one of pairs, it computes the kernel of the actual inclusion $H_2(\partial M_p) \to H_2(M_p)$, as required. $\square$

5. The distinguished hypersurface and its finite data

5.1. **The threefold.** Let $\Delta = \Delta_{\mathbb{F}_1}$ be the 22-vertex reflexive 4-polytope of [Wue26c, Theorem 6.1] and $\Delta^\vee$ its polar, with the 26 vertices $q_1, \dots, q_{26}$ listed in Table 1. The generic anticanonical hypersurface $X^\circ \subset \mathbb{P}_{\Delta^\vee}$ has, for every Δ-regular choice of equation, exactly 26 ordinary double points (one for each unit-square 2-face of $\Delta^\vee$), two anticanonical dP_7-cone points (the two pentagon faces with one interior point) and two dP_6-cone points (the two hexagon faces) [Wue26c, Section 6]. All thirty germs are *analytically* the cones: the ambient chart at the 3-dimensional cone over a 2-face F splits as $C(F) \times \mathbb{C}^*$, the restriction of the defining section to the one-dimensional orbit is $t^k(c_a + c_b t)$ with $c_a, c_b \neq 0$ by Δ-regularity because the dual edge F° has lattice length one, and Weierstrass preparation in the orbit coordinate exhibits the germ as a graph over $(C(F), 0)$. In particular each face contributes exactly one singular point, the simple zero of $c_a + c_b t$.

X° satisfies the standing hypotheses of Section 3: it is an anticanonical Cartier divisor in the Gorenstein Fano toric fourfold $P = \mathbb{P}_{\Delta^\vee}$, so $\omega_{X^\circ} \cong \mathcal{O}_{X^\circ}$ by adjunction, and $h^1(\mathcal{O}_{X^\circ}) = 0$ from the sequence $0 \to \mathcal{O}_P(K_P) \to \mathcal{O}_P \to \mathcal{O}_{X^\circ} \to 0$ together with $H^1(\mathcal{O}_P) = 0$ and $H^2(\mathcal{O}_P(K_P)) \cong H^2(\mathcal{O}_P)^\vee = 0$ (toric vanishing and Serre duality on the complete toric fourfold).

5.2. **The resolution and the matrix.** Fix the crepant resolution $\widehat{X} \to X^\circ$ obtained from the projective simplicial subdivision of the face fan of $\Delta^\vee$ using all thirty boundary lattice points, with the star subdivision at the four face-interior points and the twenty-six square

Table 1. The 30 lattice points of $\partial\Delta^\vee$: the vertices $q_1, \dots, q_{26}$ and the four face-interior points: q_{27}, q_{28} are the interior points of the two hexagons (below $E_{6,1}, E_{6,2}$) and q_{29}, q_{30} of the two pentagons (below $E_{7,1}, E_{7,2}$).

$q_1 = (-1,-1,-1,-1)$	$q_2 = (-1,-1,-1,0)$	$q_3 = (-1,-1,0,-1)$	$q_4 = (-1,-1,0,1)$
$q_5 = (-1,-1,1,0)$	$q_6 = (-1,-1,1,1)$	$q_7 = (-1,0,-1,-1)$	$q_8 = (-1,0,-1,0)$
$q_9 = (-1,0,0,-1)$	$q_{10} = (-1,0,0,1)$	$q_{11} = (-1,0,1,0)$	$q_{12} = (-1,0,1,1)$
$q_{13} = (0,-1,-1,-1)$	$q_{14} = (0,-1,-1,0)$	$q_{15} = (0,-1,0,1)$	$q_{16} = (0,-1,0,0)$
$q_{17} = (0,1,-1,-1)$	$q_{18} = (0,1,-1,0)$	$q_{19} = (0,1,0,-1)$	$q_{20} = (0,1,0,0)$
$q_{21} = (1,1,-1,-1)$	$q_{22} = (1,0,-1,0)$	$q_{23} = (0,0,0,-1)$	$q_{24} = (0,0,0,1)$
$q_{25} = (0,0,1,0)$	$q_{26} = (0,0,1,1)$	$q_{27} = (-1,-1,0,0)$	$q_{28} = (-1,0,0,0)$
$q_{29} = (0,0,-1,-1)$	$q_{30} = (0,0,-1,0)$		

diagonals

$$(q_2,q_7),(q_2,q_{13}),(q_3,q_7),(q_4,q_8),(q_4,q_{14}),(q_5,q_9),(q_5,q_{13}),(q_5,q_{23}),(q_9,q_{23}),$$
$$(q_6,q_{10}),(q_{10},q_{15}),(q_6,q_{11}),(q_6,q_{16}),(q_6,q_{25}),(q_8,q_{17}),(q_9,q_{17}),(q_{12},q_{18}),(q_{12},q_{24}),$$
$$(q_{12},q_{19}),(q_{12},q_{25}),(q_{14},q_{16}),(q_{16},q_{23}),(q_{16},q_{26}),(q_{18},q_{19}),(q_{20},q_{24}),(q_{20},q_{25}),$$

labeled $N_1,\dots,N_{26}$ in this order; the subdivision is smooth and projective (certified). Over the nodes π is small with exceptional curves Γ_i, and over the cone points the exceptional divisors are toric surfaces $E_{6,1},E_{6,2}\cong \mathrm{dP}_6$ and $E_{7,1},E_{7,2}\cong \mathrm{dP}_7$.

We record the two families of functionals on $\mathrm{Pic}(Y_{\widehat{\Sigma}})$ (equivalently, by restriction, on $\mathrm{Pic}(\widehat{X})\otimes\mathbb{Q}$, though only one containment is used below) whose kernel computation is the finite input. *Node rows*: for each square with cyclic vertices (a,b,c,d) and chosen diagonal (a,c), the intersection functional of Γ_i, with values -1 at D_a,D_c, $+1$ at D_b,D_d and 0 otherwise; it annihilates principal divisors because $q_a+q_c=q_b+q_d$. *Divisorial rows*: for each exceptional surface, the composition of the restriction $\mathrm{Pic}(Y)\to\mathrm{Pic}(E)$ ($D_j\mapsto$ the boundary curve of E for the rays of the face, $D_{\mathrm{int}}\mapsto K_E$, all other rays $\mapsto 0$) with a basis of the rank-2 or 3 lattice $K^\perp\subset\mathrm{Pic}(E)^\vee$. In the bases of Table 2 the three dP_6 functionals $\varphi_{s_1},\varphi_{s_2},\varphi_{s_3}$ satisfy $\varphi_{s_i}^2=-2$, $\varphi_{s_1}\cdot\varphi_{s_2}=1$, $\varphi_{s_3}\perp\varphi_{s_1},\varphi_{s_2}$, so $\langle\varphi_{s_3}\rangle$ is the A_1 summand and $\langle\varphi_{s_1},\varphi_{s_2}\rangle$ the A_2 summand; the two dP_7 functionals satisfy $\varphi_\beta^2=-2$, $\varphi_\alpha^2=-4$, and φ_β spans the unique (-2)-line. Stacking the 26 node rows and the ten divisorial rows gives an integral matrix $B\in\mathrm{Mat}_{36\times 26}(\mathbb{Z})$ in the dual basis of the 26 divisor classes complementary to the unimodular cone $(q_{15},q_{22},q_{24},q_{26})$.

5.3. The certified kernel computation.

Write coordinates on $\mathbb{Q}^{36}$ as

$$(u_1,\dots,u_{26};\ s_1,s_2,s_3 \text{ at } E_{6,1};\ s_1,s_2,s_3 \text{ at } E_{6,2};\ \alpha_1,\beta_1;\ \alpha_2,\beta_2).$$

The following statements are exact integral computations, reproduced independently by the ancillary programs from the printed data of Tables 1 and 2 together with the subdivision data document (see Appendix A):

(1) $\operatorname{rank} B=21$ and $\dim\ker(B^\mathsf{T})=15$; the node subsystem has rank 18, with coloops exactly N_9 and N_{11}; the unique row coloop of the full matrix is the β_1-row, i.e. β_1 vanishes identically on $\ker(B^\mathsf{T})$.

TABLE 2. Exceptional-surface data. Bases of $\mathrm{Pic}(E)$: boundary divisors at the rays listed; intersection matrices I; canonical classes K; and the $K^\perp$ functionals in the dual bases. The two dP_6 surfaces carry identical data at rays (q_3,q_4,q_5,q_6), resp. $(q_9,q_{10},q_{11},q_{12})$.

surface	lattice data	$K^\perp$ rows
$E_{6,i}$	$I_6=\begin{pmatrix}-1&0&1&0\\0&-1&0&1\\1&0&-1&1\\0&1&1&-1\end{pmatrix}$, $K=(-1,-1,-2,-2)$	$\varphi_{s_1}=(0,0,-1,1)$ $\varphi_{s_2}=(-1,-1,1,0)$ $\varphi_{s_3}=(-1,1,1,-1)$
$E_{7,1}$	$I_{7,1}=\begin{pmatrix}0&0&1\\0&-1&1\\1&1&0\end{pmatrix}$ at (q_{13},q_{17},q_{21}), $K=(-1,-1,-2)$	$\varphi_\alpha=(1,-1,0)$ $\varphi_\beta=(1,1,-1)$
$E_{7,2}$	$I_{7,2}=\begin{pmatrix}-1&0&1\\0&0&1\\1&1&0\end{pmatrix}$ at (q_{14},q_{18},q_{22}), $K=(-1,-1,-2)$	$\varphi_\alpha=(-2,0,1)$ $\varphi_\beta=(-1,-1,1)$

(2) On the *enlarged restricted kernel* $\ker(B^{\mathsf{T}}) \cap \{\alpha_1 = \alpha_2 = 0\}$, of dimension 13, the four coordinates $u_9, u_{11}, \beta_1, \beta_2$ vanish identically. The four finer profiles obtained by additionally restricting each dP_6 block to its A_1 line ($s_1 = s_2 = 0$) or its A_2 plane ($s_3 = 0$) have dimensions $9, 10, 10, 12$ and force the same four coordinates (and one s_3 on each mixed profile).

(3) The forcing is critical in the dP_7 data: releasing either α-coordinate destroys it.

By construction $\ker(B^{\mathsf{T}})$ contains the kernel of $\bigoplus_p H_2(L_p;\mathbb{Q}) \to H_2(\widehat{X};\mathbb{Q})$ written in these coordinates: the node generators map to $\pm[\Gamma_i]$ and the divisorial classes to their pushforwards from $K^\perp \subset H_2(E_p)$, and each row is the pairing with an ambient divisor class. Only this containment is used: if the ambient Picard group under-detected $H_2(\widehat{X};\mathbb{Q})$, the certified kernel would be larger than the topological one and every forced-zero statement would still bind. (In fact equality holds: $(-K_Y)^4 = 94 \neq 0$ and $h^{1,1}(\widehat{X}) = 26$; but the proofs below do not consume this.)

6. Proofs of the main results

Theorem 6.1 (mixed necessity)**.** *Let X be a compact complex threefold with $\omega_X \cong \mathcal{O}_X$ and $h^1(\mathcal{O}_X) = 0$ whose singular points are ordinary double points and anticanonical cones over dP_6 or dP_7, let $\widehat{X} \to X$ be a crepant resolution as in Section 1, and suppose X admits a one-parameter smoothing. With $R_p = \ker(H_2(L_p;\mathbb{Q}) \to H_2(M_p;\mathbb{Q}))$ for the actual Milnor fibers of the induced local smoothings, set*

$$K(X) = \Big\{\kappa \in \ker\Big(\bigoplus_p H_2(L_p;\mathbb{Q}) \to H_2(\widehat{X};\mathbb{Q})\Big) \;:\; \kappa_p \in R_p \text{ at every cone point } p\Big\}.$$

Then:

(1) *for every node q there is $\kappa \in K(X)$ with $\kappa_q \neq 0$;*

(2) *for every dP_7 point q there is $\kappa \in K(X)$ whose q-component is a nonzero element of the (-2)-line R_q;*

(3) *for every dP_6 point q smoothed through the one-parameter component of its miniversal base there is $\kappa \in K(X)$ whose q-component is a nonzero element of R_q (the A_1 summand).*

Proof. Fix small $s \neq 0$. (1) By Lemma 3.1, $\delta_q \neq 0$ in $H_3(X_s;\mathbb{Q})$; by Poincaré duality on the closed oriented X_s choose γ with $\gamma \cdot \delta_q \neq 0$; by Lemma 4.1, $\kappa = \partial\gamma$ lies in $K(X)$ and its q-component is a nonzero multiple of $\gamma \cdot \delta_q$. (2),(3) By Lemma 3.3 the generator ν of $H_3(M_q;\mathbb{Q})$ survives in $H_3(X_s;\mathbb{Q})$; choose γ with $\gamma \cdot \nu_* \neq 0$; by Lemma 4.1 the q-component of $\partial\gamma$ is the image of a nonzero element of $H_3(M_q, L_q;\mathbb{Q})$ under the isomorphism onto R_q, and R_q is identified with the stated root subspace by Proposition 2.1, Section 2.4 and Lemma 4.2. ☐

Remark 6.2. Since a $\mathbb{Q}$-vector space is not the union of finitely many proper subspaces, the per-point conclusions of Theorem 6.1 combine: there is a *single* class $\kappa \in K(X)$ simultaneously nonzero at every node, every dP_7 point, and every line-smoothed dP_6 point.

Corollary 6.3. *In the purely nodal case, Theorem 6.1(1) with Remark 6.2 states: any one-parameter smoothing forces a relation among the classes $[\Gamma_i] \in H_2(\widehat{X};\mathbb{Q})$ of the exceptional curves of a small resolution in which every node appears with nonzero coefficient.* ☐

Proposition 6.4. *If X° admitted a one-parameter smoothing, the vanishing spheres of the nodes N_9 and N_{11} would satisfy $\delta_9 = \delta_{11} = 0$ in $H_3(X_s;\mathbb{Q})$.*

Proof. By Lemma 4.1 every boundary class $\partial\gamma$ lies in $K(X^\circ)$; its dP_7-components lie on the (-2)-lines by Proposition 2.1, and Lemma 4.2 identifies these with the (-2)-lines of the resolution-side coordinates of Section 5.3, so $\partial\gamma$ lies in the enlarged restricted kernel, on which u_9 and u_{11} vanish identically (item (2) there). Since the u_q-component of $\partial\gamma$ is a nonzero multiple of $\gamma \cdot \delta_q$, nondegeneracy of the intersection pairing gives $\delta_9 = \delta_{11} = 0$. $\square$

Theorem 6.5. *X° admits no one-parameter smoothing.*

Proof. Suppose one exists. Every germ of X° is then smoothed; in particular the node N_9 is, so Lemma 3.1 gives $\delta_9 \neq 0$ in $H_3(X_s; \mathbb{Q})$, contradicting Proposition 6.4.

Alternatively, avoiding the nodes altogether: the point covered by $E_{7,1}$ is smoothed through the unique component of its miniversal base, so Theorem 6.1(2) produces $\kappa \in K(X^\circ)$ whose $E_{7,1}$-component is a nonzero element of the (-2)-line, i.e. (Lemma 4.2 and Table 2) has nonzero β_1-coordinate; but β_1 vanishes identically already on the unrestricted $\ker(B^\mathsf{T})$ (Section 5.3, item (1)). $\square$

Remark 6.6. The two proofs are independent in the period lemma and in the certified fact consumed (Lemma 3.1 with Section 5.3(2), versus Lemma 3.3 with Section 5.3(1)), and share the membership and boundary-compatibility inputs (Lemmas 4.1 and 4.2) and the divisorial rows of Table 2.

Corollary 6.7. *No fiber of a small representative of the miniversal deformation of X° is smooth; consequently X° is not smoothable over any base germ, and neither is any small deformation of X°. Likewise, no formal arc in the miniversal deformation of X° meets all thirty local smoothing loci.*

Proof. The miniversal deformation of the compact X° exists [Gra74] and its discriminant is a closed analytic subset of the base. If some fiber arbitrarily close to the origin were smooth, the curve selection lemma [Mil68] would produce a real-analytic arc through the origin with smooth generic fiber; complexifying (a real-analytic function vanishing on a real segment of a disk vanishes identically, and the discriminant preimage of the complexified arc is discrete) gives a holomorphic disk, that is, a one-parameter smoothing, contradicting Theorem 6.5. Small deformations are covered by openness of versality. For the formal statement: by Artin approximation [Art68] applied to the analytic miniversal deformation, a formal arc whose local components avoid all thirty local discriminants would be approximated, to any jet order beyond the vanishing orders of those components, by an analytic arc with the same property. Since the singular locus of the fibers is closed in the total space and meets the central fiber only in the thirty points, the fibers of the approximating arc are, for small t, singular at most inside the thirty Milnor-ball neighborhoods; avoidance of the local discriminants makes them smooth there, hence smooth everywhere, so the arc is a one-parameter smoothing, contradicting Theorem 6.5. (The local components are read through choices of local classifying maps; avoidance of the local discriminants is choice-independent, being smoothness of the local fibers.) $\square$

Remark 6.8. The mirror partner of X° in the doubly isolated pair carries the anticanonical cone over $\mathbb{F}_1$, which admits no local smoothing [Wue26a]. Thus both members of the unique mixed pair are non-smoothable, one locally and one globally, although the thirty germs of X° are each smoothable. The two nodal coloops quantify how far X° is from the nodal theory: no relation among the twenty-six exceptional curves alone is nonzero at N_9 or N_{11}, so a smoothing would have had to route through the divisorial directions, and items (2)–(3) of Section 5.3 say the divisorial directions cannot supply what is needed at any order.

7. The criterion as a finite test, and an open problem

Theorem 6.1 is a finite test for any concrete candidate: compute the matrix of Section 5.2 from the polytope (node circuit rows; $K^\perp$ bases of the star fans of the pentagon and hexagon faces), restrict to the branch subspaces, and check whether some covered coordinate is forced to vanish. A forced node coordinate contradicts Lemma 3.1; a forced dP_7 root coordinate contradicts Lemma 3.3; a forced s_3-coordinate contradicts Lemma 3.3 on profiles through the line component. The test consumes nothing about the example beyond Δ-regularity and its face inventory.

We ran the test on the anticanonical hypersurface X_{19} of the 19-vertex reflexive polytope Δ_{19} printed in [Wue26c, Section 5.3], whose generic member has fourteen nodes and one dP_7-cone point, all isolated. The certified outcome is structurally opposite to X°:

(1) the node subsystem has rank 12 with two coloops: as for X°, no purely nodal mechanism can smooth X_{19};
(2) on the full kernel (dimension 3) the complementary dP_7-coordinate α is already forced to zero, so every relation class lies on the (-2)-line without imposing the branch restriction; and
(3) no covered coordinate is forced: a generic kernel class is nonzero in all fourteen node coordinates and in the (-2)-coordinate simultaneously.

The necessity theory of this paper is therefore silent on X_{19}, which motivates Question 1.5. As a control in the opposite direction, the test applied to the generic one-nodal quintic threefold is silent as well, as it must be: there the exceptional curve class vanishes in H_2 of the small resolution (the threefold is $\mathbb{Q}$-factorial), the kernel is the whole line $H_2(L;\mathbb{Q})$, and the node coordinate is realized, the smoothable side of the nodal picture. A smoothing, if it exists, must recruit the dP_7 direction; the natural construction tools are ambient toric degenerations in the sense of Ilten–Vollmert [IV12], and we leave the question to future work.

8. Discussion

8.1. **The two-parameter degree-6 channel.** Theorem 6.1 covers dP_6 points smoothed through the one-parameter component. For the two-parameter (plane) component the Milnor fiber has $b_3 = 2$, and the argument of Lemma 3.3 reduces the analogous nonvanishing to the $\mathbb{Q}$-linear independence of two periods of Ω_M on $W \setminus \bar{S}$, a finer question than nonvanishing, which the toric method above does not decide. None of the results of this paper require it.

8.2. **First-order theory.** The tangent-level analogue of the test of Section 5.3 identifies $\ker(B^\mathsf{T})$ with the image of the localization map on first-order deformations, by the local-to-global comparison of Friedman–Laza [FL25, FL24]; from that viewpoint the content of this paper is that the first-order forced-zero statements persist at every order of contact, which no first-order argument can deliver. The necessity statements of [Fri86, RT09] are of first-order type, with the upgrade to actual smoothings supplied in the nodal case by unobstructedness; the present method needs neither, which is what allows it to reach the non-lci cone points.

8.3. **Sufficiency.** The converse direction, constructing mixed smoothings when the test is silent, is open even for X_{19}. The purely nodal comparison is instructive: in dimension three the converse of Friedman's criterion holds for projective nodal Calabi–Yau threefolds, by the first-order relation together with the unobstructedness theorems of Kawamata and Tian, while in higher odd dimensions Rollenske and Thomas prove a converse for nodal hypersurfaces in projective space and explain why a general converse should not be expected

[RT09, Section 5]. We record one structural observation from the toric side: for a face F with $\ell(F^\circ) = 1$ the polar vertices adjacent to F give ambient deformation degrees whose slice polytopes are Minkowski-rigid, so any ambient construction must use degrees on the extension of the dual edge, the exact combinatorial locus of Altmann's local smoothing parameter. This will be taken up elsewhere.

Appendix A. Computational certificates

The exact lattice and matrix claims used in the proofs are reproduced by two short, self-contained Python programs in exact rational arithmetic, independently of any computer-algebra system. A third self-contained program combines exact integral checks with a floating-point diagnostic of the period parametrization, whose underlying identity is proved exactly in Lemma 3.2. The general test program reuses the companion repository's exact toric source modules. The inputs are Tables 1 and 2, the ancillary subdivision and restriction data documents, and the polytope of [Wue26c, Section 5.3]:

- the lattice package (`milnor_kernels.py`): the boundary structure of the exceptional surfaces, the norms and pairings of the $K^\perp$ rows, the root decompositions, the (-2)-enumerations with rigorous search bounds, and the isometries to the standard del Pezzo markings;
- the matrix package (`global_kernel.py`): the reconstruction of all 36 rows from the printed data, the ranks, coloops, kernels, the four profiles and the enlarged profile with their forced-zero lists, and the criticality statement of Section 5.3(3);
- the period package (`dp_periods.py`): the exact fan combinatorics behind (2) (the ± 1 pairings on all rays, the divisor computation, the slice self-intersection patterns), the corner-chart transitions, and the constancy of the pullback in Lemma 3.2, verified to machine precision at random points;
- the repository-integrated test package (`examples.py`): the general implementation of Section 7, validated end-to-end by reproducing every number of Section 5.3 for X° from scratch, and the Δ_{19} computation.

The fixed subdivision of Section 5.2 and its smoothness and projectivity are certified by an exact integral support function; the Δ-regularity of the reference equation of X°, including nondegeneracy along every face, is certified at an integral witness (`resolution_fan.sage` and `global_section.sage`, with five further Sage certificates and the face-data module `fixed_example.py`).

The arXiv ancillary bundle includes this directory together with the shared exact toric modules imported by the repository-integrated programs. The same files are public, together with the programs and data of the companion papers, at

`https://github.com/bwuebben/calabi-yau-smoothability`.

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